

Day 8 — Project Session 1: Build the Rāśi-Wheel

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a substantial draft of the *rāśi*-wheel project — labelled, coloured, with the four solstice/equinox markers and at least 6 of the 12 *saṁkrānti* dates filled in.

Materials

- The in-progress *rāśi*-wheel from Day 5 (one per pair)
- Coloured pencils, sketch pens, markers
- A4 / A3 chart paper for an optional larger wheel
- Glue, scissors, ruler
- The *sources.md* file printed for reference
- Project rubric (*assessments/rubric.md*) — printed and visible

Lesson flow

Warm-up (5 min)

- Recap the project brief (5 sentences): each pair builds a *rāśi*-wheel that shows BOTH the *sāyana* and *nirayana* dates for the four turning-points, the 12 *rāśi* names with English/Greek equivalents, and at least one paragraph linking it to axial tilt and seasons.
- Show the rubric (it should already be on the board). Quick Q&A: 2 minutes for any rubric clarifications.

Core (5 min)

The teacher explicitly walks through the “Meets” target for the project:

- 12 *rāśi* names in correct order ✓
- English / Greek equivalents matching ✓
- Approximate *nirayana* date ranges (e.g. Meṣa = 14 Apr – 14 May) ✓
- All four solstice / equinox markers labelled with both *sāyana* and *nirayana* dates ✓
- A 4-sentence written explanation of why we have seasons (axial tilt) ✓
- Citation to *Sūrya-siddhānta* OR *Bṛhat Saṁhitā* OR an NCERT chapter ✓

To **Exceed**, add ONE of: - A clearly-drawn precession diagram showing the wobble cone - A second wheel for Greek-origin star-name comparisons - A short paragraph on the cross-cultural Sanskrit-Greek-Babylonian zodiac correspondence

Build time (30 min)

Pairs work in silence first (10 min) to get the labels down, then collaborative (20 min) to add the explanatory text, colour, and diagrams.

Teacher circulates:

- For pairs stuck on labelling: hand them the printed reference table from Day 5.
- For pairs stuck on axial-tilt explanation: redirect to the Day 2 sketch in their notebook.
- For pairs racing ahead: ask “what would happen if the tilt were 0°?” or “how would a Babylonian astronomer label this wheel differently?”

Check-in (5 min)

Stop work at minute 40. Each pair holds up their current draft. The teacher does a quick visual scan and notes 2–3 pairs to circle back to first thing tomorrow.

- Exit prompt: “*What’s the ONE thing you still need to finish before tomorrow?*”
- Write it on a sticky note attached to the wheel.

Pair coaching prompts (for the teacher)

If you see ...	Try saying ...
One partner doing all the drawing	“Tell me what your partner is contributing. Where will their writing go?”
A wheel with English names only	“Where do the Sanskrit names go? Show me.”
Confusion about <i>sāyana</i> vs <i>nirayana</i>	“Pick the four turning-points. Write the December solstice date on one line. Now write the Makara Saṁkrānti date below it. They’re different — your job is to make that visible on the wheel.”
Beautiful art, no astronomy	“Where on the wheel does the reader learn that the seasons come from axial tilt? Add a sentence.”

Differentiation

- **One tier up:** Add a second concentric ring showing the Western horoscope (*sāyana*) date range for each *rāśi*. The reader should be able to compare the two rings at a glance.
- **One tier down:** Drop the *sāyana* / *nirayana* dual labelling. Just label one set of dates clearly. The *ayanāmśa* concept appears only in the written paragraph.

Teacher notes

- Some pairs will spend the entire 45 minutes colouring and lose the astronomy. Intervene at minute 20: “show me your written paragraph on axial tilt.” If it doesn’t exist yet, stop the colouring.
- A surprising number of pairs will get the *rāśi* order slightly wrong (especially *Tulā* ↔ *Vṛścika*). Catch this early.

Homework

- Finish at home what didn’t get done in class. Bring back tomorrow ready to refine, not start.

- If the pair worked separately, agree (over WhatsApp or in person) who finishes what.